Shuigen Liu

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Positions

2026 Aug – **Postdoctoral Fellow**, The Pennsylvania State University
Mentor: Prof. John Harlim

2025 – 2026 **Research Fellow**, National University of Singapore

2024 Jul – Dec **Visiting student**, Massachusetts Institute of Technology

Education

2021 – 2025 **Ph.D. in Mathematics**, National University of Singapore
Advisors: Prof. Xin T. Tong and Prof. Weizhu Bao

2017 – 2021 **B.S. in Mathematics**, Peking University

Research Interests

My research lies at the intersection of computational mathematics, applied probability and statistics. I focus on analysis and development of algorithms for high-dimensional sampling and uncertainty quantification, with applications in Bayesian inference, inverse problems and generative modeling. I am particularly interested in localization methods for sampling. I also work on ensemble Kalman methods for high-dimensional inference and data assimilation.

Keywords: Sampling, uncertainty quantification, localization, dimension reduction, data assimilation.

Publications and Preprints

Preprints

[6] Georg A. Gottwald, **Shuigen Liu**, Youssef Marzouk, Sebastian Reich, and Xin T. Tong, *Localized diffusion models*, arXiv:2505.04417, submitted to Bernoulli (2026).

[5] Tiangan Cui, **Shuigen Liu**, and Xin T. Tong, *Stein's method for marginals on large graphical models*, arXiv:2410.11771, submitted to Bernoulli (2026).

Published/accepted

[4] **Shuigen Liu**, and Xin T. Tong, *Spectral Gap of Metropolis Algorithms for Non-smooth Distributions under Isoperimetry*, Jpn. J. Stat. Data Sci. (2026).

[3] Ningxin Liu, **Shuigen Liu**, Xin T. Tong, and Lijian Jiang, *Estimate of Koopman modes and eigenvalues with Kalman filter*, SIAM J. Sci. Comput., 48 (2026), pp. A512-A539.

[2] Rafael Flock, **Shuigen Liu**, Yiqiu Dong, and Xin T. Tong, *Local MALA-within-Gibbs for Bayesian image deblurring with total variation prior*, SIAM J. Sci. Comput., 47 (2025), pp. A2127–A2153.

[1] **Shuigen Liu**, Sebastian Reich and Xin T. Tong, *Dropout ensemble Kalman inversion for high dimensional inverse problems*, SIAM J. Numer. Anal., 63 (2025), pp. 685–715.

In preparation

Shuigen Liu, Youssef Marzouk, Aimee Maurais, and Xin T. Tong, *Sampling by Localized Flows*, (2026+).

Tiangang Cui, **Shuigen Liu**, Xin T. Tong, Romain Verdiere, and Olivier Zahm *Localized Triangular Transport Map*, (2026+).

Teaching

National University of Singapore

TA	Finite Element Method, Spring 2025
TA	Essential Data Analytics Tools: Numerical Computation, Spring 2024
TA	Calculus for Computing, Spring 2023 and Fall 2023
TA	Essential Data Analytics Tools: Convex Optimization, Spring 2023

Awards

2024	Faculty of Science PhD Conference Award, National University of Singapore
2024	Overseas Research Immersion Award, National University of Singapore
2021	President's Graduate Fellowship, National University of Singapore
2021	Excellent Graduate, Peking University

Service

Reviewer for: SIAM Journal on Numerical Analysis, SIAM/ASA Journal on Uncertainty Quantification, Journal of Machine Learning Research, Statistics and Computing, Japanese Journal of Statistics and Data Science

Conferences

Jul 2025	Efficient Sampling Algorithms for Complex Models, IMS NUS, Singapore.
Jun 2025	MATRIX Research Program: Bayesian Learning of Very High-Dimensional Physical Process Models, MATRIX, Melbourne, Australia.
May 2025	2025 STU-NUS PhD Forum, Shanghai Jiao Tong University, Shanghai, China.
Oct 2024	FDS Conference: Recent Advances and Future Directions for Sampling, Yale University, New Haven, U.S.
Feb 2024	2024 SIAM Conference on Uncertainty Quantification (UQ24), Trieste, Italy.
Aug 2023	2023 International Congress on Industrial and Applied Mathematics (ICIAM 2023), Waseda University, Tokyo, Japan.
Jun 2023	Foundations of Computational Mathematics 2023 Conference (FoCM 2023), Sorbonne University, Paris, France.
Jul 2022	2022 International Conference on Scientific Computation and Differential Equations (SciCADE 2022), University of Iceland, Reykjavík, Iceland.